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An Engine-Regulated Covering Law for Little Red Dots: One Equilibrium Relation for Their Spectral Sequence, Demography, and Broad-Line Widths

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ABSTRACT

Little red dots (LRDs) present three puzzles usually treated separately: an ordered spectral-subtype sequence, a rise–peak–decline number density, and broad Balmer lines in compact, X-ray-weak sources. Within a semi-analytic lifecycle model with a dense-plus-diffuse soft-EUV emission layer, we show that all three respond to a single equilibrium relation: the covering odds of the nuclear reprocessor equal the square of the intrinsic engine-to-host continuum contrast, $C/(1-C) = x^2$ with $x \equiv L_{\nu,\mathrm{engine}}(2500\,\text{Å})/L_{\nu,\star}(2500\,\text{Å})$ and unit normalization. Confronted directly with the eight stacked [O III]/Hβ and [O III]/[O II] ratios of the four continuum subtypes, the raw ionizing-photon partition under this law—supplemented by one physically motivated geometric parameter, the stellar ionizing coupling $\epsilon \simeq 0.1$, consistent with canonical Lyman-continuum escape fractions—matches the data at least as well as an imposed photon-partition closure ($\chi^2 = 10.4$ versus 12.9 on the identical metric; seed range 5–16), while the demographic fit on the reliable $z \leq 7.5$ bins improves from $\chi^2 = 6.9$ to 1.2 and the RUBIES broad-width distribution distance improves from 0.41 to 0.35 with the widths reinterpreted as substantially electron-scattering broadened ($\tau_\mathrm{e} \simeq 2.3$). The exponent is not fitted but constructed and then confronted: of four candidate mechanisms, roll-over (sequential-stage) and steepened (Poisson double-blocking) laws are rejected by the data, cooling-fractionation is excluded both by its saturated timescales and by the demonstrated metallicity-blindness of the law ($\Delta\chi^2 > 120$ against any Z-modulation), leaving a unique survivor among the candidates considered—momentum-limited envelope building against drag-limited clump removal—whose precondition we verify holds for 100% of the selected population, and whose structure identifies $\gamma = 2$ with the number of soft modes of the doubly forced clump system. The model predicts a subtype "march" (selected red fraction 0.04 ± 0.05 at $z \simeq 9.5$ rising to 0.49 ± 0.09 at $z \simeq 4.5$) and blue LRDs that are more X-ray transmitting and more variable than red LRDs at matched luminosity. We also show the reported $z \simeq 9.5$ abundance cannot currently constrain formation: transparent variations of one color-selection gate change the predicted density by an order of magnitude while moving the $z \leq 7.5$ bins by well under their errors.

Keywords: Active galactic nuclei (16) — Supermassive black holes (1663) — High-redshift galaxies (734) — Galaxy evolution (594)

1. INTRODUCTION

JWST has revealed an abundant population of compact, red, frequently broad-line sources at $z \gtrsim 4$, the little red dots (Matthee et al. 2024; Greene et al. 2024; Kocevski et al. 2024; Akins et al. 2024; Kocevski et al. 2025). Three empirical regularities frame the present work. First, stacked spectroscopy resolves an ordered continuum-subtype sequence, from extreme red (xLRD) through blue (bLRD) classes, with common line species but strongly varying line-to-continuum structure (Pérez-González et al. 2026), a diversity also mapped to black-hole luminosity and host contrast in a stationary framework (Barro et al. 2025). Second, broad-selection number densities rise, peak near $z \sim 5$–6, and decline toward both lower and higher redshift (Rinaldi et al. 2026; Ma et al. 2026). Third, the population combines broad Balmer lines in very compact regions with pronounced X-ray

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weakness (Ananna et al. 2024; Yue et al. 2024; Maiolino et al. 2024; Sneppen et al. 2026) and generally low photometric variability (Tee et al. 2024; Secunda et al. 2025; Kokubo & Harikane 2025, the last for a five-object sample), plausibly indicating dense, Compton-thick circumnuclear gas. A wave of recent models interprets LRDs as black holes embedded in dense gas envelopes with quasi-stellar photospheres and scattering-broadened Balmer lines (Naidu et al. 2025; Rusakov et al. 2025; de Graaff et al. 2025; Sneppen et al. 2026b), a programme since extended to unified "black-hole-star" families and explicit nuclear-minus-host decompositions (de Graaff et al. 2025b; Sun et al. 2026) and to quasi-star evolutionary tracks and late-stage quasi-star models (Santarelli et al. 2025; Begelman & Dexter 2026; Kido et al. 2025); super-Eddington accretion states offer an alternative account of the X-ray weakness (Pacucci & Narayan 2024; Inayoshi et al. 2025), though deep Chandra limits challenge some variants (e.g., Sacchi & Bogdán 2025); see the current X-ray census of Tortosa et al. (2026). Published clustering measurements additionally bound the host halos (Pizzati et al. 2025; Carranza-Escudero et al. 2025; Wang et al. 2026), a constraint the lifecycle's halo occupation must ultimately face. Our model shares the dense-envelope architecture and the low-spin entry channel proposed for LRD demographics by Pacucci & Loeb (2025); its distinguishing content is a population-level equilibrium law connecting the envelope to the host, and the joint confrontation of that law with the subtype sequence, the demography, and the width distribution simultaneously.

In a companion demographic analysis (Pandev 2026a, in preparation) we concluded that binned number densities alone cannot select a formation mechanism. Here we take the opposite approach: a forward semi-analytic lifecycle model whose observation layer is confronted simultaneously with the stacked spectral sequence, the number densities, public broad-line widths (Hviding et al. 2025), and population-level hardness constraints (Sok et al. 2026). The central result is that these independent confrontations select, and are jointly improved by, one simple relation between the covering of the nuclear reprocessor and the visibility of the nuclear engine—and that the relation's form can be constructed from an explicit elimination program rather than fitted.

Section 2 summarizes the model. Section 3 develops the covering law and its derivation status. Section 4 presents the confrontations, Section 5 the predictions, Section 6 the status of the high-redshift bin, and Section 7 the limitations.

2. MODEL

2.1. *Lifecycle*

The population model evolves 1.2×10^4 halo histories, importance reweighted to the Tinker et al. (2008) mass function, with mean growth from Fakhouri et al. (2010) and halo bias from Tinker et al. (2010). Each halo carries nuclear gas, black-hole, and burst-reservoir masses through the states quiescent → compacting → embedded → clearing → post-LRD, with recurrence. Entry depends on spin (Bullock et al. 2001), specific growth, and nuclear Thomson depth (low-angular-momentum compaction; Mo et al. 1998); embedded objects exit through competing gas-exhaustion, AGN-feedback, and host/size-growth hazards. Population survival is emergent: no survival kernel is imposed. A secondary gas-rich burst entry channel operates in rapidly assembling high-redshift hosts; it transfers a finite baryon reservoir and is mass-conserving. One global abundance normalization is calibrated at $4.5 < z < 6.5$ and is refit for each model configuration; comparative statistics below therefore compare shapes, not absolute abundances.

2.2. *Observation layer*

The porous-envelope layer assigns each active source a two-scale geometry: a cool reprocessor of outer scale 1360 AU with covering fraction C, clump-population UV/X-ray transfer, and a compact line-emitting scale calibrated in Section 4.3. The forward continuum is the sum of an attenuated stellar host, a leaked accretion component, and the reprocessor photosphere; sources are classified into the four continuum subtypes by the published L_{5100}/L_{2500} boundaries (1.8, 3.1, 6.3 from bLRD through xLRD), and the "V-shape color" below denotes the model's rest-frame optical-redness proxy used by its selection function (the empirical V-shaped turnover at the Balmer limit is characterized by Setton et al. 2024). The emission layer is a dense-plus-diffuse two-zone photoionization model built on thermal-balance Cloudy anchors (C23 release; a locally verified C25 build is used; Chatzikos et al. 2023): a dense, dust-free skin producing H/He lines with collisionally suppressed forbidden lines, and a diffuse skin producing the optical metal lines, mixed at the level of photon yields by the dense fraction of intercepted ionizing photons, $f_{\rm dense}$. Public stacks reject a standard AGN ionizing continuum through He II: the data require a soft basis that preserves 35–54 eV photons while suppressing flux above the He^+ edge; a ten-screen warm-absorber scan shows a generic ionized filter cannot achieve this, so a soft basis (labelled by a 50 kK spectral shape, not a literal temperature) is adopted for all sources (Pérez-González et al. 2026; Sok et al. 2026).

3. THE COVERING LAW AND ITS DERIVATION STATUS

3.1. *Decomposition of the imposed closure*

Inverting the two-zone model through the four stacked subtype spectra yields an ordered sequence in the dense fraction of processed ionizing photons: 0.981, 0.901, 0.738, and 0.086 (xLRD→bLRD; intervals in Table 2). We emphasize that these are model-conditional inversions—obtained through the same two-zone emission layer used forward—not direct measurements; the directly observed quantities are the stacked line ratios confronted in Section 4.1. Defining the nuclear-to-host continuum dominance $D \equiv L_{\nu,\mathrm{thermal}}(5100)/L_{\nu,\mathrm{host}}(5100)$, the inverted sequence is reproduced by the quadratic closure $f_{\mathrm{dense}} = D^2/(1+D^2)$, with a free exponent returning 2.02 (conditional on a declared 0.05 systematic floor discussed in Section 4.1; 1.89 with measurement errors alone).

To determine what supplies the two powers, we constructed every physical photon partition available upstream of the closure—envelope-intercepted engine photons against diffuse illumination by stellar, escaped-nuclear, and dust-quenched supplies—and measured the emergent source-level exponent of $\log(Q_{\mathrm{dense}}/Q_{\mathrm{diffuse}})$ on $\log D$ (Figure 1). Restricted to the physically populated range $|\log_{10} D| < 1.5$ (the extended panels also contain numerically capped sources at extreme D, produced by a floor on the host term; they are excluded from all fits and flagged in the caption), the fiducial candidates give $\eta = 0.65 \pm 0.10$ and 0.66 ± 0.11; the legacy Thomson-depth covering contributes 0.09 powers (slope of covering odds on $\log D$). Ordinary flux competition therefore supplies roughly one of the two required powers, and the second must come from co-regulation of the interception itself. We record for completeness that our pre-declared expectation—that installing a covering law would raise the emergent source-level exponent to $\simeq 2$—was not borne out (it rises only to 0.65–0.89); the resolution is that the stacked observables constrain subtype-level medians, not the source-level slope, and the within-subtype scatter of Section 5 is the observable that tests the difference.

3.2. *The covering law*

We therefore replace the Thomson-depth covering with an engine-regulated law. With $x \equiv L_{\nu,\mathrm{engine}}(2500\,\text{Å})/L_{\nu,\star}(2500\,\text{Å})$ the *intrinsic* (pre-attenuation, pre-covering) contrast, the covering odds are

$$\frac{C}{1-C} = k\,x^{\gamma}, \qquad (1)$$

with a frozen per-halo lognormal scatter of 0.35 dex (this value is an assumption: a scan over 0.20–0.50 dex changes the spectral statistic by $< 5\%$, so the scatter is not constrained by the present observables). Both terms in x are computed upstream of the observables inside the simulator; we caution that this does not by itself make the calibration non-circular, and we address circularity by construction: the law's parameters are assessed against the directly observed line ratios (Section 4.1) and the $z \leq 7.5$ demography, and the model-conditional inversions of Section 3.1 are used only as diagnostics. A scan over $\gamma \in \{1.5, 2, 2.5, 3\}$, $k \in \{0.5, 1, 1.5, 2\}$ (a descriptive sum of the spectral and demographic statistics, not a likelihood) prefers $(\gamma, k) = (2, 1)$; the preference is stable across four random seeds and against removal of the contested $z = 9.5$ bin (Section 6), and under the fiducial line-ratio-space objective the nearest competitor—the $\gamma = 3$, $k = 0.5$ point—sits at joint $\simeq 32$ versus 11.5, a factor $\sim$3 worse; the tabulated $k = 1$ profile is itself non-monotonic in γ (Table 1), reflecting selection reshuffling between laws. Covering odds equal the squared contrast with 50% covering at engine–host parity.

3.3. *Equilibrium structure, anchors, and bounds*

Three features of the law are derivable. (i) *The odds form.* If covering evolves by competition between a building rate per uncovered solid angle, B, and a clearing rate per covered solid angle, G, $\dot{C} = (1-C)B - CG$ equilibrates exactly at $C/(1-C) = B/G$. (ii) *The normalization.* Identifying B with the engine momentum flux and G with host feedback, the ratio at a shared interface is radius-independent, $B/G \sim (L_e/c)/(\dot{p}_{\mathrm{SN}}\,\mathrm{SFR})$ with $\dot{p}_{\mathrm{SN}} \simeq 3 \times 10^{33}$ dyn per $M_\odot\,\mathrm{yr}^{-1}$ (Kim & Ostriker 2015), and evaluates to $\simeq 3$ at $x = 1$: the anchor argument therefore fixes k to order unity within a factor ~ 3, while the scan discriminates at the factor-2 level; we present the anchor as a consistency check at its actual precision, not as a derivation of $k = 1$. (iii) *Fast relaxation.* A time-resolved implementation (exact exponential-integrator update; covering initialized unbuilt at episode entry) shows the stationary law is the fast-relaxation limit: relaxation timescales of 30, 80, and 300 Myr degrade the spectral statistic monotonically (defining the bound by where the degradation exceeds the seed scatter gives $\tau_{\mathrm{relax}} \lesssim 30$ Myr, the shortest value resolvable at the 12 Myr integration step). The failure mode is fossil covering: envelopes built while the engine was strong and retained as it fades repopulate the blue subtype class with dense sources. The bound sits six orders of magnitude above the $\sim$10 yr dynamical time at

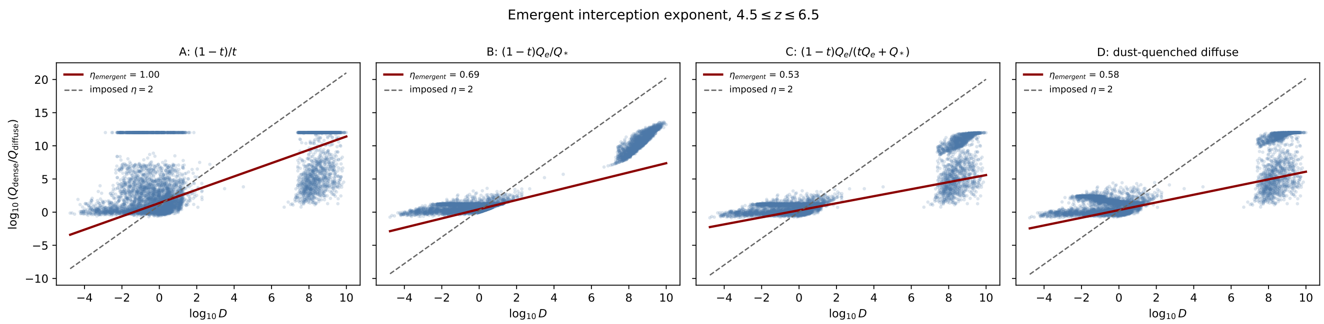

Figure 1. Emergent source-level exponent of four physical photon partitions against the continuum dominance D at $4.5 \le z \le 6.5$. Fits (red, with standard errors) are restricted to the physically populated range $|\log_{10} D| < 1.5$ (unshaded); the shaded regions contain numerically capped sources (floors on the host term and yield caps) excluded from all fits. Ordinary flux competition supplies roughly one power of D; the second must come from the covering law.

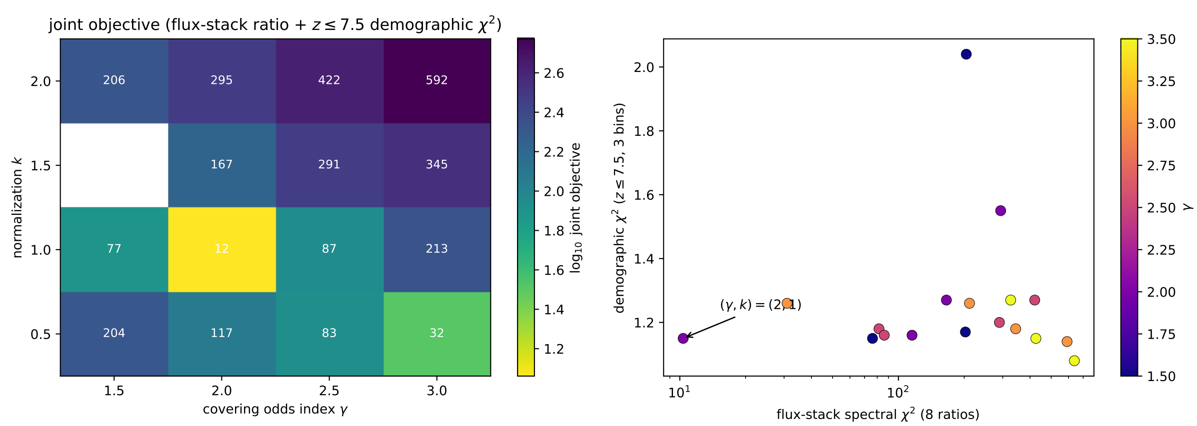

Figure 2. Joint scan of the covering law under the fiducial metric. Left: joint objective (flux-stacked line-ratio χ^2 plus $z \le 7.5$ demographic χ^2; logarithmic color scale) over the evaluated (γ, k) cells; the preferred point is the unit-normalization quadratic. Right: spectral versus demographic components for all evaluated points. The objective is a descriptive sum; no probabilistic reading of Δ values is intended.

the envelope scale, so fast equilibration is the expected regime.

3.4. *The exponent: a completed elimination program*

The mass budget alone cannot supply the second power of x: converting the engine's clump supply and the host's ablation flux to solid-angle rates, the clump surface density cancels between build and clear, so any mass-limited build against entrainment-limited clearing yields exactly one power of x regardless of confinement details. Four candidate mechanisms for the second power were formulated and confronted (Table 1):

Route B (sequential two-stage odds), $C/(1-C) = (kx)^2/(1+2kx)$ with the combinatorial structure forcing $k \simeq 1$: rejected in both forms under the fiducial metric (Table 1). The parameter-free form fails catastrophically because its high-contrast roll-over flattens the law across the populated range ($x \simeq 0.5$–2); the profile minimum in k ($k = 3$, which would also displace the per-stage parity to $x = 1/3$) still loses to the pure quadratic by a margin an order of magnitude beyond the seed scatter.

Route D (Poisson double-blocking), $C = 1 - e^{-kx}(1 + kx)$ from marginally translucent clumps requiring two-deep sightlines: rejected at every normalization; its exponential steepening beyond $kx \sim 2$ distorts the subtype spacing in the opposite direction.

Route A (cooling-versus-mixing fractionation): eliminated twice over. Theoretically, at the model's own radiation-confinement pressures the cooling times at the build interface are 10^{-2}–10^{0} yr across the plausible parameter space, while macroscopic clumps mix over decades: the fractionation is saturated by 2–5 orders of magnitude and cannot carry a power of x (cf. cloud-crushing and survival scalings; Klein et al. 1994; Gronke & Oh 2018). Empirically, Route A's unavoidable fingerprint—covering odds proportional to the cooling function, hence to metallicity—is rejected: multiplying the odds by $(Z/0.2\,Z_\odot)^\delta$ and rerunning the full model gives $\Delta\chi^2 > 120$ for each tested modulation ($\delta = -0.5$, $+0.5$, $+1$; Table 1); the law is metallicity-blind at the tested amplitudes.

Route C (momentum-limited build against drag-limited removal): the unique survivor, with its precondition verified rather than assumed. The engine's momentum-driven lofting capacity $L_e/(cv_w)$ is 14–29% (p16–p84) of the available nuclear mass supply, below unity for 100% of the selected population in every subtype: the build bottleneck is driving, not mass, which bypasses the surface-density cancellation. The removal channel is likewise selected by the physics: the same fast cooling that saturates Route A makes thermal clump destruction ineffective (mixed gas recondenses), so clumps exit the interception zone by drag entrainment, whose timescale carries the standard density-contrast scaling $t_{\rm drag} \propto \chi\, r/v$ with $\chi \propto P_{\rm rad}/P_{\rm host} \propto x$. The chain

$$\frac{C}{1-C} = \frac{B}{G} \propto \frac{L_e/(cv_w)}{1/t_{\rm drag}} \propto x \cdot x = x^2 \qquad (2)$$

then has every structural property the confrontations demand: scale-freeness (required by the B and D rejections), metallicity blindness, the parity anchor, and fast relaxation. In its strongest form the exponent is combinatorial: it counts the soft modes of the doubly forced clump system—population (build/clear) and residence (confinement/drag)—each equilibrating linearly in the field ratio, with opacity frozen ($\tau \gg 1$; a soft opacity mode would imprint the rejected Route-D curvature) and stage survival not independent (its distinctness would imprint the rejected Route-B roll-over). The confrontations select this count among the candidates considered: the scan disfavors $\gamma = 1.5$ and $\gamma = 3$ alongside the B and D functional forms (a selection under a descriptive objective, not a probabilistic measurement). The one imported ingredient is the χ^1 drag-time scaling, standard in the cloud-acceleration literature; a targeted simulation of a radiation-confined clump in a host-driven wind remains the external anchor.

Table 1. The elimination program for the covering-law exponent. All spectral statistics are the flux-stacked line-ratio χ^2 over eight observables under the identical fiducial metric of Section 4.1 ($\epsilon = 0.1$); joint adds the $z \leq 7.5$ demographic term.

candidate law	spectral χ^2	joint
pure x^2 (γ=2, k=1)	10.4	11.5
$\gamma = 1.5$ (k=1)	76.1	77.3
$\gamma = 2.5$ (k=1)	86.2	87.3
$\gamma = 3$ (k=1)	211.4	212.7
Route B, k=1 (parameter-free)	165.1	166.6
Route B, best k (k=3)	27.3	28.9
Route D, best k (k=1.68)	51.9	53.2
metallicity-modulated, δ=+1	132.9	135.8
metallicity-modulated, δ=−0.5	241.1	246.7

Note—The $k = 1$ profile in γ is non-monotonic (76.1, 10.4, 86.2, 211.4 at $\gamma = 1.5$, 2, 2.5, 3), with a minimum at $\gamma = 2$: changing the law reshuffles which sources are selected into each subtype, so the objective is not a smooth function of the exponent. The preference for $(2, 1)$ over every alternative nevertheless exceeds the seed scatter by an order of magnitude.

4. JOINT CONFRONTATIONS

Statistical conventions. We report unreduced χ^2 sums under a declared error model and do not convert them to likelihoods, p-values, or $\Delta\chi^2$ probabilities; the covering-law scan objective in particular is a descriptive sum of the spectral and demographic terms (Section 3.2), and parameters are selected on coarse grids without propagated posteriors (Section 7). The spectral statistic sums $N = 8$ stacked observables—the [O III]λ5007/Hβ and [O III]/[O II] ratios of the four continuum subtypes—against the published stack errors, with three parameters selected on this metric (γ, k, and the stellar ionizing coupling ϵ), leaving 5 residual degrees of freedom ($\chi^2_\nu \simeq 2.1$). The demographic statistic sums the three $z \leq 7.5$ bins against their published errors (no cosmic-variance term) with one refit abundance normalization, leaving 2 residual degrees of freedom ($\chi^2_\nu \simeq 0.6$). The joint objective is their unweighted sum ($N = 11$ observables, 4 selected parameters). Reduced values are indicative only.

4.1. *The stacked line ratios*

The decisive spectral test is performed directly in observable space. For each subtype we form the luminosity-weighted stack of per-source line yields (mirroring how observed stacks average spectra) and compare the resulting [O III]λ5007/Hβ and [O III]/[O II] ratios with the eight stacked measurements. One physical ingredient beyond Equation 1 is required: the fraction ϵ of surviving stellar ionizing photons that reaches the nuclear-scale diffuse gas. Diagnosing the raw partition shows the xLRD diffuse budget would otherwise be dominated by stellar leak-through at a level six times too high, while no (γ, k) can compensate (an 18-point grid fails uniformly in ratio space with $\epsilon = 1$). Physically, most stellar Lyman-continuum photons are consumed in local H II regions; the canonical escape fractions of 5–20% (e.g., Flury et al. 2022) bracket the fitted $\epsilon = 0.10$ (profile flat over 0.05–0.15), so we identify ϵ with the LyC escape fraction into the wider ISM and treat its agreement as a consistency check rather than a free success. We state plainly that ϵ was diagnosed on, and fitted to, the same eight ratios it now helps match; it is counted among the three selected parameters in Table 2. The comparison value for the imposed closure (12.9) is evaluated on the identical catalogue and metric; the closure sets $f_{\rm dense}$ directly from D and therefore has no ϵ degree of freedom (its own selected parameters are the exponent and the systematics floor of its original calibration). Because the seed range of the physical partition (5.0–15.6) straddles 12.9, "at least as well" is a fiducial-seed statement.

Table 2 shows the result: the raw photon partition under the fitted covering law matches the stacked ratios at least as well as the imposed quadratic closure evaluated on the identical metric ($\chi^2 = 10.4$ versus 12.9), and passes the independent He II gate. Two disclosures accompany this. First, in the earlier inversion-space metric the imposed closure scores 0.38 versus the physical partition's 1.6; that metric's 0.05 systematic floor—20 times the xLRD statistical interval—controls those values, which is why we adopt the direct ratio-space confrontation as fiducial. Second, the bLRD central value is matched by neither model and its interval does not discriminate; headline agreement claims rest on the three red subtypes.

4.2. *Demography*

On the three $z \leq 7.5$ bins (the $z \simeq 9.5$ bin is selection-dominated; Section 6), the demographic χ^2 improves from 6.90 (Thomson-depth covering) to 1.15 under the fitted law, with one refit normalization and no covering parameter tuned on this statistic alone. Two caveats: the $z = 9.5$ pull flips sign between configurations ($+2.3 \rightarrow -2.1\sigma$), partly because the burst-channel rate was originally tuned under the superseded covering model (a 6–60 Gyr^{-1} rate scan leaves all conclusions unchanged); and the quoted bin errors are the published ones, without an added cosmic variance term. The model's selected xLRD fraction at the anchor redshift (0.2% under the UV gate) remains a factor $\sim$60 below the observed stack-sample share, a selection-layer failure reported in the closure analysis and unchanged here. We also note that the demographic shape itself is contested: a recent analysis reports no evolution in the LRD number density from cosmic dawn to cosmic noon (Loiacono et al. 2026), in tension with the rise–peak–decline bins fitted here, while a spectroscopic census across $3 < z < 6$ (Zhuang et al. 2025) and an analysis of the $z < 3$ decline (Zhang et al. 2026) further underscore that the demographic shape is observationally unsettled—a disagreement consistent with our conclusion (Section 6) that current LRD demography is dominated by selection calibration.

4.3. *Broad-line widths*

Rerunning the public RUBIES width confrontation (Hviding et al. 2025) under the fitted law improves the distribution-level agreement (weighted CDF distance 0.354 versus 0.413; a descriptive statistic—no null distribution is available—computed against broad-component

Table 2. Direct confrontation with the stacked line ratios at $4.5 \leq z \leq 6.5$ under the fitted law with $\epsilon = 0.1$. The inverted dense fractions (final columns) are shown as model-conditional diagnostics only.

subtype	[O III]/Hβ model	observed	$f_{\rm dense}$ model	inverted (p16–p84)
xLRD	1.20	1.10 ± 0.10	0.95	0.981 (0.978–0.983)
plusLRD	3.00	3.20 ± 0.10	0.91	0.901 (0.884–0.911)
minusLRD	3.98	4.40 ± 0.30	0.75	0.738 (0.625–0.807)
bLRD	4.59	4.90 ± 0.50	0.45	0.086 (0.000–0.658)

Note—Total $\chi^2 = 10.4$ over the eight [O III]/Hβ and [O III]/[O II] observables (seed range 5.0–15.6), with three selected parameters (γ, k, ϵ); the imposed D^2 closure evaluated on the identical catalogue and metric gives 12.9. All four He IIλ4686/Hβ predictions are $\leq$ 0.015, below the population gate of 0.1. The bLRD inverted interval is unconstrained and its central value is not matched; only the three red subtypes discriminate.

fits that were censored at 2500 km s^{-1}) and changes its interpretation (Figure 3): the matched population shifts to higher black-hole masses (median $\log M_{\mathrm{BH}}$ 6.85→7.36) and Compton-thick Thomson depths ($\tau_{\mathrm{e}} \simeq 2.3$), so $\simeq 1650$ km s^{-1} of the width budget is electron-scattering broadening—consistent with envelope models in which scattering shapes the Balmer profiles (Naidu et al. 2025; de Graaff et al. 2025)—and the descriptive compact scale moves from 1.2×10^3 to 5.9×10^3 AU (bootstrap 5.5–7.1×10^3; the scan support was widened from the published 5000 AU edge to keep this fit interior, which we disclose as a boundary effect of the original support). The factor-of-five scale shift between covering models quantifies a systematic on any compact-scale inference. We note an unresolved geometric tension: the scattering envelope's calibrated outer scale (1360 AU) lies inside the descriptive line-emitting scale, whereas a scatterer must overlie its emitter; the single-scale envelope is evidently oversimplified, and an extended scattering halo ($\gtrsim 6 \times 10^3$ AU) is required for self-consistency (Section 7). The headline scattering quantities survive this relocation: τ_{e} is computed from the nuclear gas column, not from the 1360 AU photosphere, so the $\simeq 1650$ km s^{-1} budget is unchanged; what the relocation does alter is the covering geometry of the scattering region, which is no longer constrained by the continuum-reprocessor calibration—an uncertainty we estimate at the tens-of-percent level on the width distribution shape.

Separately, the Hα calibration marker—the offset between the model's selected-population median Hα luminosity and that of a UV- and redshift-matched public JADES control sample (not an LRD sample)—moves from +0.89 to +0.64 dex; a factor-4 residual remains and this quantity is a consistency indicator, not a fit.

5. PREDICTIONS

(i) The subtype march. Because D grows as reservoirs feed while hosts assemble, the subtype mixture must evolve. Seed-averaged, with combined binomial and seed uncertainties, the selected red fraction (xLRD+plusLRD) is 0.04 ± 0.05, 0.09 ± 0.04, 0.23 ± 0.05, 0.49 ± 0.09, and 0.48 ± 0.14 at $z \simeq 9.5$, 7.5, 5.5, 4.5, and 3.25, while the bLRD fraction falls from 0.61 ± 0.15 to $\lesssim 0.05$ (Figure 4). Removing the selection weighting strengthens the trend, so a transparent gate cannot manufacture it; the rise is monotonic in all four seed realizations for $z \geq 4.5$. The predicted slope between $z \simeq 7.5$ and 4.5 is 0.10–0.14 per unit redshift. A full statement of this prediction, its uncertainties, its exposure to data-side selection evolution, and its falsification thresholds is given in the companion Letter (Pandev 2026c), which this section summarizes. The recent discovery of two LRDs transitioning into quasars (Fu et al. 2025) provides direct evidence for evolutionary transitions at the phase exit, complementary to the within-phase march predicted here.

(ii) X-ray and variability contrast. Blue LRDs carry unbuilt envelopes, so at matched bolometric luminosity they should be systematically more X-ray transmitting and more variable than red LRDs, with a steepened variability gradient across the subtype sequence. Only the ordering is claimed; published LRD monitoring finds low variability overall, with tentative variability in a minority of sources in larger multi-epoch samples (Tee et al. 2024; Secunda et al. 2025; Kokubo & Harikane 2025), and the prediction concerns the differential between subtypes at matched luminosity, which existing samples have not yet tested.

(iii) Within-subtype scatter. A deterministic D^2 law predicts a tight line-ratio locus at fixed continuum class; the covering chain predicts scatter from the covering log-normal. Because the scatter amplitude is unconstrained by present data (Section 3.2), the prediction is a range; object-level [O III]/Hβ dispersion within subtypes discriminates the two pictures.

(iv) Saturation homogeneity and no hysteresis. Covering saturates (reaches the 0.995 cap) near $x \simeq 14$, so the reddest sources should be homogeneous in covering-driven marks; and no thick loop should appear in the variability–color plane—the fast-equilibration hierarchy of Section 3.3 forbids it.

(v) Scale-free granularity. The soft-mode construction of Section 3.4 is self-similar: covering-driven fluctuations (absorption variability, X-ray transmission granularity in partially covered sources) should be scale-free; a measured characteristic clump size would falsify the framework independently of the lifecycle.

6. THE $Z \simeq 9.5$ BIN IS A COMPLETENESS THERMOMETER

The fitted model underpredicts the reported $z \simeq 9.5$ broad-selection density by a factor 6.3, a residual robust to the burst-rate scan. Its character is resolved by a dichotomy test on the forward catalogue. The intrinsic $M_{\mathrm{UV}} < -18.5$ nuclear-active population at $z \simeq 9.5$ exceeds the reported density by a factor $\sim$34 (a model-conditional, order-of-magnitude statement: the estimate reweights catalogued sources by their selection probabilities, and its headroom resides in sources with $p_{\mathrm{sel}} = 10^{-6}$–$10^{-3}$ assigned by an unvalidated analytic gate). These sources are UV-bright, compact, and embedded, and fail selection only through the V-shape color gate, because at this epoch the covering law leaves most envelopes unbuilt and continua blue. Rescaling

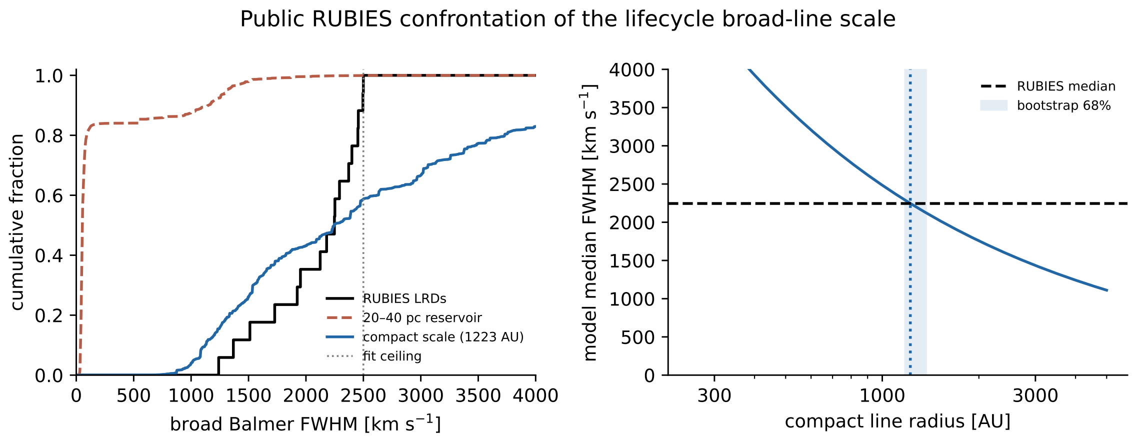

Figure 3. RUBIES broad Balmer width confrontation under the fitted covering law. Left: cumulative width distributions; the dotted vertical line marks the 2500 km s^{-1} ceiling imposed in the published broad-component fits (the model CDFs are not truncated). The dashed curve shows the superseded 20–40 pc reservoir-scale width model for reference. Right: descriptive compact-scale calibration; the model median matches the observed median at 5.9×10^3 AU with a substantial electron-scattering contribution.

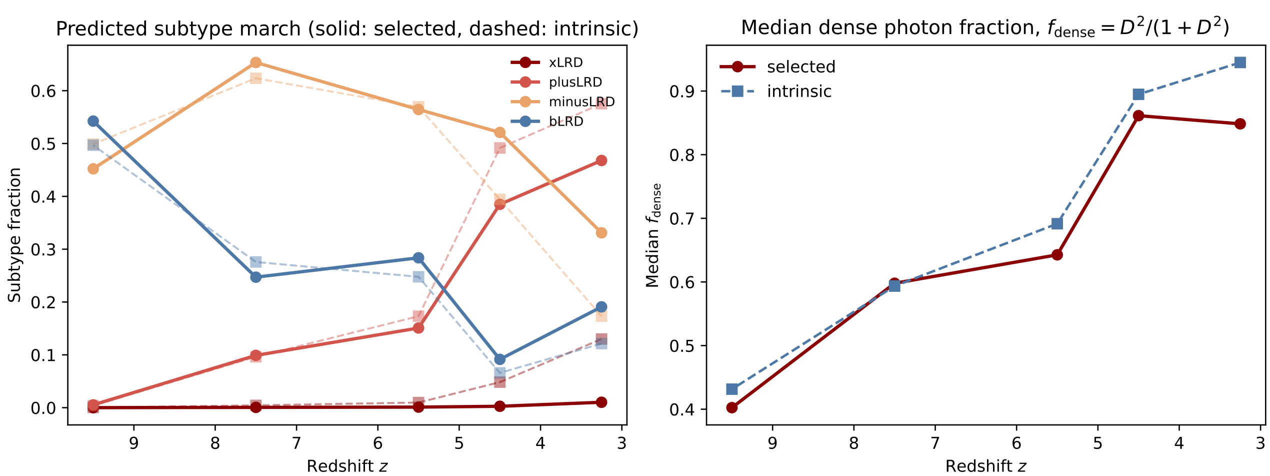

Figure 4. The predicted subtype march (single fiducial realization; the seed-averaged fractions with combined binomial-plus-seed uncertainties are given in the text and in the companion Letter). Left: subtype fractions versus redshift, selection-weighted (solid) and with selection removed (dashed); the trend strengthens without selection. Right: median dense photon fraction. The monotonic rise of the red fraction is established across seeds for $z \gtrsim 4.5$; the $z \simeq 3.25$ bin (effective sample $\simeq$19) is an extrapolation below the calibration range and is plateau-consistent.

that single gate within transparent limits ($\sim$0.4 mag in the redness threshold) moves the predicted $z = 9.5$ density from a factor $\sim$6 deficit to a factor $\sim$3 excess of the observed value while moving the $z \leq 7.5$ bins by well under their errors. Three conclusions follow. No additional formation channel is required by current number densities. The $z \simeq 9.5$ bin cannot presently discriminate formation physics; it measures the survey's color completeness, which Rinaldi et al. (2026) describe as essentially unconstrained at $z > 6$–7 (their discussion of MIRI depth). And the falsifiable content is compositional: the real $z > 8.5$ LRD population should be dominated by blue-continuum, low-covering compact sources (quantified in the Letter, including the per-subtype gate-passing fractions that reconcile bLRD dominance of the selected sample with the removal of the blue headroom); a red-dominated census would falsify this resolution. The decisive external products are injection/recovery completeness for blue compact templates and the effective-area masks.

7. LIMITATIONS

The observation layer remains a set of transparent proxies: the selection gates are not the survey pipelines; the X-ray transfer over-suppresses known bands; and the Balmer decrement is not solved (intrinsic Hα/H$\beta \simeq 2.8$–3.2 against observed 5.5–10.1 (Pérez-González et al. 2026); a smooth screen with $A_V \simeq 2$–4 toward the line region is not rejected). The covering-law parameters are selected on coarse grids without propagated posteriors; the covering scatter is assumed; ϵ is fitted (its agreement with canonical escape fractions is a consistency, not a derivation); and the drag-time scaling of Section 3.4 is imported from standard cloud physics. The single-scale scattering envelope is internally inconsistent with the descriptive line-emitting scale and requires an extended halo. The published RUBIES widths are censored at 2500 km s^{-1}, so the compact-scale calibration is descriptive. Demographic errors omit cosmic variance. Seed statements rest on four realizations. None of the fits here is a posterior; a catalogue-level likelihood with injection/recovery completeness remains the requirement for population inference.

8. SUMMARY

A single equilibrium relation—covering odds equal to the squared intrinsic engine-to-host contrast, with a momentum-parity normalization—simultaneously (1) matches the stacked line-ratio sequence through a raw photon partition at least as well as an imposed closure, given one geometric parameter consistent with canonical LyC escape fractions, (2) improves the reliable-bin demographic fit to $\chi^2 = 1.2$ on 3 bins, (3) improves and reinterprets the public broad-width confrontation in favor of electron-scattering broadening, and (4) converts the high-redshift abundance residual into a quantified statement about color completeness. The exponent is constructed, not fitted: of four candidate mechanisms confronted, three were eliminated by the data (with the eliminations consistent with the clump system having exactly two soft modes), and the surviving candidate's precondition—momentum-limited building—holds for the entire selected population. The framework's failure modes—a red-dominated $z > 8.5$ census, a tight within-subtype line-ratio locus, an inverted subtype march, a thick variability–color hysteresis loop, or a measured characteristic clump scale—are all measurable with data expected on a 1–2 year horizon.

This work is unfunded and was carried out independently. It made use of public data products from the JWST archive and of published catalogues by the teams cited herein; the author thanks those teams for making their data and measurements public. This research made use of NASA's Astrophysics Data System and the arXiv preprint repository.

Facilities: JWST (NIRSpec, NIRCam, MIRI), Chandra (data limits as cited)

DATA AND CODE AVAILABILITY

The lifecycle model, covering-law scans, Cloudy anchor grids, stack inversion, elimination-program scripts, and all confrontation scripts accompany this paper as an MIT-licensed code archive, together with machine-readable results for every table and figure (Zenodo DOI to be assigned at submission); the full 65-configuration forward-catalogue grid is deposited as a separate CC-BY-4.0 data record (Zenodo DOI to be assigned). Only public data products are used: the Rinaldi et al. (2026) and Ma et al. (2026) binned densities, the Pérez-González et al. (2026) stacks, the Hviding et al. (2025) RUBIES release, the Sok et al. (2026) hardness constraint, and public JADES catalogues (Eisenstein et al. 2023).

Software: Cloudy (Chatzikos et al. 2023), numpy, scipy, matplotlib, astropy, colossus

REFERENCES

Akins, H. B., et al. 2024, arXiv:2406.10341

Ananna, T. T., et al. 2024, ApJL, 969, L18

Barro, G., et al. 2025, arXiv:2512.15853

Begelman, M. C., & Dexter, J. 2026, ApJ, 996, 48

Bullock, J. S., Dekel, A., Kolatt, T. S., et al. 2001, ApJ, 555, 240

Carranza-Escudero, M., et al. 2025, ApJL, 989, L50

Chatzikos, M., et al. 2023, RMxAA, 59, 271

de Graaff, A., et al. 2025, arXiv:2503.16600

de Graaff, A., et al. 2025, arXiv:2511.21820

Eisenstein, D. J., et al. 2023, arXiv:2306.02465

Fakhouri, O., Ma, C.-P., & Boylan-Kolchin, M. 2010, MNRAS, 406, 2267

Flury, S. R., et al. 2022, ApJS, 260, 1

Fu, S., et al. 2025, arXiv:2512.02096

Greene, J. E., et al. 2024, ApJ, 964, 39

Gronke, M., & Oh, S. P. 2018, MNRAS, 480, L111

Hviding, R. E., et al. 2025, RUBIES broad-line data release, Zenodo, doi:10.5281/zenodo.16758549

Kido, D., et al. 2025, MNRAS, 544, 3407

Kim, C.-G., & Ostriker, E. C. 2015, ApJ, 802, 99

Klein, R. I., McKee, C. F., & Colella, P. 1994, ApJ, 420, 213

Kocevski, D. D., et al. 2024, arXiv:2404.03576

Kocevski, D. D., et al. 2025, ApJ, 986, 126

Inayoshi, K., Kimura, S. S., & Noda, H. 2025, PASJ, 77, 811

Kokubo, M., & Harikane, Y. 2025, ApJ, 995, 24

Loiacono, F., et al. 2026, arXiv:2606.30253

Ma, Y., et al. 2026, ApJ, 1000, 59

Maiolino, R., et al. 2024, arXiv:2405.00504

Matthee, J., et al. 2024, ApJ, 963, 129

Mo, H. J., Mao, S., & White, S. D. M. 1998, MNRAS, 295, 319

Naidu, R. P., et al. 2025, arXiv:2503.16596

Pacucci, F., & Loeb, A. 2025, ApJL, arXiv:2506.03244

Pacucci, F., & Narayan, R. 2024, ApJ, arXiv:2407.15915

Pérez-González, P. G., et al. 2026, arXiv:2602.20247

Pizzati, E., et al. 2025, MNRAS, 539, 2910

Rinaldi, P., et al. 2026, arXiv:2604.07138

Rusakov, V., et al. 2025, Nature, arXiv:2503.16595

Sacchi, A., & Bogdán, Á. 2025, ApJL, 989, L30

Santarelli, A. D., et al. 2025, arXiv:2510.17952

Secunda, A., et al. 2025, arXiv:2509.03571

Setton, D. J., et al. 2024, arXiv:2411.03424

Sneppen, A., et al. 2026, arXiv:2605.15263

Sneppen, A., et al. 2026, arXiv:2601.18864

Sok, V., et al. 2026, arXiv:2606.23778

Sun, W. Q., et al. 2026, The Open Journal of Astrophysics, 9

Tee, W. L., et al. 2024, arXiv:2412.05242

Tinker, J., et al. 2008, ApJ, 688, 709

Tinker, J., et al. 2010, ApJ, 724, 878

Tortosa, A., et al. 2026, A&A, 708, A293

Wang, Z., et al. 2026, arXiv:2603.15736

Yue, M., et al. 2024, arXiv:2404.13290

Zhang, C., et al. 2026, arXiv:2606.02773

Zhuang, M.-Y., et al. 2025, arXiv:2505.20393